■ CYBERNOVA TECHNOLOGIES
Active Directory Capstone Project

Complete Step-by-Step Implementation Guide

Enterprise Identity and Access Management (IAM) Deployment

Domain	cybernova.local
Platform	VMware / Azure (Windows Server 2022)
Submission Deadline	14th June, 2026
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■■ PHASE 1: VIRTUAL LAB SETUP

Set up your virtual environment before anything else. You need 1 server and at least 1 client machine.

1.1 VMware VM Configuration

VM	OS	RAM	Storage	Role
DC01	Windows Server 2022	4 GB	60 GB	Domain Controller
CLIENT01	Windows 10/11	2 GB	40 GB	Client Machine 1
CLIENT02	Windows 10/11	2 GB	40 GB	Client Machine 2

1.2 Network Settings

- Set all VMs to the SAME network adapter — use Host-Only or NAT with same subnet
- Assign a STATIC IP to the Domain Controller

Static IP configuration for DC01 (Server):

```
IP Address: 192.168.1.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1
Preferred DNS: 127.0.0.1 (points to itself after AD install)
```

■ Take a screenshot of the network adapter settings as evidence.

■■ PHASE 2: INSTALL ACTIVE DIRECTORY DOMAIN SERVICES

2.1 Rename the Server

Before installing AD DS, rename your server to DC01:

```
# Open PowerShell as Administrator Rename-Computer -NewName 'DC01' -Restart
```

■ After restart, log back in and proceed.

2.2 Install AD DS Role via PowerShell

```
Install-WindowsFeature -Name AD-Domain-Services -IncludeManagementTools
```

2.3 Promote Server to Domain Controller

```
Install-ADDSForest ` -DomainName "cybernova.local" ` -DomainNetbiosName "CYBERNOVA" `
-InstallDns:$true ` -SafeModeAdministratorPassword (ConvertTo-SecureString
"Admin@12345" -AsPlainText -Force) ` -Force:$true
```

■■ Server will automatically RESTART after promotion. This is normal.

■ After reboot, log in as CYBERNOVA\Administrator

2.4 Verify AD Installation

```
# Confirm AD DS is running Get-ADDomain # Check domain controller
Get-ADDomainController
```

■ Screenshot the output of Get-ADDomain as proof of setup.

■ PHASE 3: CONFIGURE ORGANIZATIONAL UNITS (OUs)

OUs allow you to organize users, computers, and resources by department. This mirrors real enterprise structure.

3.1 Create All OUs via PowerShell

```
# Create top-level OUs New-ADOrganizationalUnit -Name "IT Department" -Path
"DC=cybernova,DC=local" New-ADOrganizationalUnit -Name "HR Department" -Path
"DC=cybernova,DC=local" New-ADOrganizationalUnit -Name "Finance Department" -Path
"DC=cybernova,DC=local" New-ADOrganizationalUnit -Name "Remote Users" -Path
"DC=cybernova,DC=local" # Create sub-OUs under each department New-ADOrganizationalUnit
-Name "Computers" -Path "OU=IT Department,DC=cybernova,DC=local"
New-ADOrganizationalUnit -Name "Users" -Path "OU=IT Department,DC=cybernova,DC=local"
New-ADOrganizationalUnit -Name "Users" -Path "OU=HR Department,DC=cybernova,DC=local"
New-ADOrganizationalUnit -Name "Users" -Path "OU=Finance
Department,DC=cybernova,DC=local" New-ADOrganizationalUnit -Name "Users" -Path
"OU=Remote Users,DC=cybernova,DC=local"
```

3.2 Verify OUs Were Created

```
Get-ADOrganizationalUnit -Filter * | Select-Object Name, DistinguishedName
```

■ Open Active Directory Users and Computers (ADUC) GUI and take a screenshot showing the OU tree.

3.3 OU Design Justification (for your report)

OU	Purpose	Justification
IT Department	IT admins and computers	Need elevated privileges and separate GPOs
HR Department	HR staff accounts	Handle sensitive data; restricted access needed
Finance Department	Finance staff	Financial data protection; strict access control
Remote Users	VPN/remote workers	Different logon/policy requirements

■ PHASE 4: CREATE USERS AND GROUPS

4.1 Create Security Groups

```
New-ADGroup -Name "IT_Admins" -GroupScope Global -GroupCategory Security ` -Path "OU=IT
Department,DC=cybernova,DC=local" ` -Description "IT Administrators Group" New-ADGroup
-Name "HR_Staff" -GroupScope Global -GroupCategory Security ` -Path "OU=HR
Department,DC=cybernova,DC=local" ` -Description "HR Department Staff" New-ADGroup
-Name "Finance_Staff" -GroupScope Global -GroupCategory Security ` -Path "OU=Finance
Department,DC=cybernova,DC=local" ` -Description "Finance Department Staff" New-ADGroup
-Name "Remote_Users" -GroupScope Global -GroupCategory Security ` -Path "OU=Remote
Users,DC=cybernova,DC=local" ` -Description "Remote/VPN Users"
```

4.2 Create User Accounts (12 Users)

Use this template for each user. Replace values as needed:

```
# IT Department Users New-ADUser -Name "John Mensah" -GivenName "John" -Surname
"Mensah" ` -SamAccountName "jmensah" -UserPrincipalName "jmensah@cybernova.local" `
-Path "OU=Users,OU=IT Department,DC=cybernova,DC=local" ` -AccountPassword
(ConvertTo-SecureString "Pass@1234" -AsPlainText -Force) ` -Enabled $true
-PasswordNeverExpires $false New-ADUser -Name "Abena Owusu" -GivenName "Abena" -Surname
"Owusu" ` -SamAccountName "aowusu" -UserPrincipalName "aowusu@cybernova.local" ` -Path
"OU=Users,OU=IT Department,DC=cybernova,DC=local" ` -AccountPassword
(ConvertTo-SecureString "Pass@1234" -AsPlainText -Force) ` -Enabled $true New-ADUser
-Name "Kwame Asante" -GivenName "Kwame" -Surname "Asante" ` -SamAccountName "kasante"
-UserPrincipalName "kasante@cybernova.local" ` -Path "OU=Users,OU=IT
Department,DC=cybernova,DC=local" ` -AccountPassword (ConvertTo-SecureString
"Pass@1234" -AsPlainText -Force) ` -Enabled $true # HR Department Users New-ADUser
-Name "Ama Boateng" -GivenName "Ama" -Surname "Boateng" ` -SamAccountName "aboateng"
-UserPrincipalName "aboateng@cybernova.local" ` -Path "OU=Users,OU=HR
Department,DC=cybernova,DC=local" ` -AccountPassword (ConvertTo-SecureString
"Pass@1234" -AsPlainText -Force) ` -Enabled $true New-ADUser -Name "Kofi Darko"
-GivenName "Kofi" -Surname "Darko" ` -SamAccountName "kdarko" -UserPrincipalName
"kdarko@cybernova.local" ` -Path "OU=Users,OU=HR Department,DC=cybernova,DC=local" `
-AccountPassword (ConvertTo-SecureString "Pass@1234" -AsPlainText -Force) ` -Enabled
$true New-ADUser -Name "Efua Asare" -GivenName "Efua" -Surname "Asare" `
-SamAccountName "easare" -UserPrincipalName "easare@cybernova.local" ` -Path
"OU=Users,OU=HR Department,DC=cybernova,DC=local" ` -AccountPassword
(ConvertTo-SecureString "Pass@1234" -AsPlainText -Force) ` -Enabled $true # Finance
Department Users New-ADUser -Name "Yaw Adjei" -GivenName "Yaw" -Surname "Adjei" `
-SamAccountName "yadjei" -UserPrincipalName "yadjei@cybernova.local" ` -Path
"OU=Users,OU=Finance Department,DC=cybernova,DC=local" ` -AccountPassword
(ConvertTo-SecureString "Pass@1234" -AsPlainText -Force) ` -Enabled $true New-ADUser
-Name "Akosua Frimpong" -GivenName "Akosua" -Surname "Frimpong" ` -SamAccountName
"afrimpong" -UserPrincipalName "afrimpong@cybernova.local" ` -Path "OU=Users,OU=Finance
Department,DC=cybernova,DC=local" ` -AccountPassword (ConvertTo-SecureString
"Pass@1234" -AsPlainText -Force) ` -Enabled $true New-ADUser -Name "Nana Appiah"
-GivenName "Nana" -Surname "Appiah" ` -SamAccountName "nappiah" -UserPrincipalName
"nappiah@cybernova.local" ` -Path "OU=Users,OU=Finance
Department,DC=cybernova,DC=local" ` -AccountPassword (ConvertTo-SecureString
"Pass@1234" -AsPlainText -Force) ` -Enabled $true # Remote Users New-ADUser -Name
"Fiifi Mensah" -GivenName "Fiifi" -Surname "Mensah" ` -SamAccountName "fmensah"
-UserPrincipalName "fmensah@cybernova.local" ` -Path "OU=Users,OU=Remote
Users,DC=cybernova,DC=local" ` -AccountPassword (ConvertTo-SecureString "Pass@1234"
```

```
-AsPlainText -Force) ` -Enabled $true New-ADUser -Name "Adwoa Kusi" -GivenName "Adwoa"
-Surname "Kusi" ` -SamAccountName "akusi" -UserPrincipalName "akusi@cybernova.local" `
-Path "OU=Users,OU=Remote Users,DC=cybernova,DC=local" ` -AccountPassword
(ConvertTo-SecureString "Pass@1234" -AsPlainText -Force) ` -Enabled $true New-ADUser
-Name "Kweku Boadu" -GivenName "Kweku" -Surname "Boadu" ` -SamAccountName "kboadu"
-UserPrincipalName "kboadu@cybernova.local" ` -Path "OU=Users,OU=Remote
Users,DC=cybernova,DC=local" ` -AccountPassword (ConvertTo-SecureString "Pass@1234"
-AsPlainText -Force) ` -Enabled $true
```

4.3 Add Users to Their Groups

```
# Add IT users to IT_Admins Add-ADGroupMember -Identity "IT_Admins" -Members
"jmensah","aowusu","kasante" # Add HR users to HR_Staff Add-ADGroupMember -Identity
"HR_Staff" -Members "aboateng","kdarko","easare" # Add Finance users to Finance_Staff
Add-ADGroupMember -Identity "Finance_Staff" -Members "yadjei","afrimpong","nappiah" #
Add Remote users to Remote_Users Add-ADGroupMember -Identity "Remote_Users" -Members
"fmensah","akusi","kboadu"
```

4.4 Verify Users and Groups

```
# List all users Get-ADUser -Filter * | Select-Object Name, SamAccountName, Enabled #
List group members Get-ADGroupMember -Identity "IT_Admins" | Select-Object Name
Get-ADGroupMember -Identity "HR_Staff" | Select-Object Name Get-ADGroupMember -Identity
"Finance_Staff" | Select-Object Name Get-ADGroupMember -Identity "Remote_Users" |
Select-Object Name
```

■ Screenshot all group membership outputs as evidence.

■ PHASE 5: GROUP POLICY OBJECTS (GPOs)

GPOs enforce security settings across all machines in the domain. Open Group Policy Management Console (GPMC) or use PowerShell.

5.1 Open GPMC

```
# Open Group Policy Management Console gpmc.msc
```

5.2 Password Policy GPO

Apply at domain level. Go to: Default Domain Policy > Computer Config > Policies > Windows Settings > Security Settings > Account Policies > Password Policy

```
# Via PowerShell (Fine-Grained Password Policy) New-ADFineGrainedPasswordPolicy -Name
"CyberNova-Password-Policy" ` -Precedence 1 ` -MinPasswordLength 10 `
-PasswordHistoryCount 5 ` -ComplexityEnabled $true ` -MaxPasswordAge "30.00:00:00" `
-MinPasswordAge "1.00:00:00" ` -ReversibleEncryptionEnabled $false # Apply to Domain
Users group Add-ADFineGrainedPasswordPolicySubject -Identity
"CyberNova-Password-Policy" ` -Subjects "Domain Users"
```

5.3 Account Lockout Policy

Go to: Default Domain Policy > Account Policies > Account Lockout Policy

Setting	Value
Account lockout threshold	5 invalid attempts
Account lockout duration	30 minutes
Reset account lockout counter after	15 minutes

5.4 Restrict Control Panel Access

Create a new GPO called 'Restrict-Standard-Users' and link to HR and Finance OUs.

Path: User Config > Policies > Admin Templates > Control Panel

```
# In GPMC GUI: # Right-click HR Department OU > Create a GPO > Name:
"Restrict-Standard-Users" # Edit > User Configuration > Administrative Templates >
Control Panel # Enable: "Prohibit access to Control Panel and PC settings" # Link same
GPO to Finance OU as well
```

5.5 Restrict Command Prompt for Non-Admins

Path: User Config > Policies > Admin Templates > System

```
# Edit the same 'Restrict-Standard-Users' GPO # User Configuration > Administrative
Templates > System # Enable: "Prevent access to the command prompt" # Also enable:
"Prevent access to registry editing tools"
```

5.6 Automatic Screen Lock Policy

Path: User Config > Policies > Admin Templates > Control Panel > Personalization

```
# In GPO Editor: # User Configuration > Administrative Templates > Control Panel >
Personalization # Enable: "Enable screen saver" # Enable: "Password protect the screen
saver" # Set: "Screen saver timeout" = 600 seconds (10 minutes) # Also set via Computer
Config for stronger enforcement: # Computer Config > Windows Settings > Security
```

```
Settings > Local Policies > Security Options # "Interactive logon: Machine inactivity limit" = 600
```

5.7 Force Group Policy Update

```
# Run on Domain Controller to push GPOs immediately gpupdate /force # Run on client machines too gpupdate /force # Verify GPO application gpresult /r
```

■ Screenshot gpresult /r output on a client machine to show GPOs are applying correctly.

■ PHASE 6: FILE SHARING AND NTFS PERMISSIONS

Create shared folders for each department with strict Role-Based Access Control (RBAC).

6.1 Create Shared Folders

```
# Create base directory structure New-Item -Path "C:\Shares" -ItemType Directory
New-Item -Path "C:\Shares\IT_Share" -ItemType Directory New-Item -Path
"C:\Shares\HR_Share" -ItemType Directory New-Item -Path "C:\Shares\Finance_Share"
-ItemType Directory
```

6.2 Set Share Permissions

```
# Share IT_Share folder New-SmbShare -Name "IT_Share" -Path "C:\Shares\IT_Share" `
-FullAccess "CYBERNOVA\IT_Admins" ` -ReadAccess "CYBERNOVA\Domain Admins" # Share
HR_Share folder New-SmbShare -Name "HR_Share" -Path "C:\Shares\HR_Share" ` -FullAccess
"CYBERNOVA\HR_Staff" ` -NoAccess "CYBERNOVA\Finance_Staff","CYBERNOVA\Remote_Users" #
Share Finance_Share folder New-SmbShare -Name "Finance_Share" -Path
"C:\Shares\Finance_Share" ` -FullAccess "CYBERNOVA\Finance_Staff" ` -NoAccess
"CYBERNOVA\HR_Staff","CYBERNOVA\Remote_Users"
```

6.3 Set NTFS Permissions

```
# Set NTFS permissions on IT_Share $acl = Get-Acl "C:\Shares\IT_Share"
$acl.SetAccessRuleProtection($true, $false) $rule = New-Object
System.Security.AccessControl.FileSystemAccessRule(
"CYBERNOVA\IT_Admins","FullControl","ContainerInherit,ObjectInherit","None","Allow")
$acl.SetAccessRule($rule) Set-Acl "C:\Shares\IT_Share" $acl # Repeat for HR_Share with
HR_Staff group $acl2 = Get-Acl "C:\Shares\HR_Share"
$acl2.SetAccessRuleProtection($true, $false) $rule2 = New-Object
System.Security.AccessControl.FileSystemAccessRule(
"CYBERNOVA\HR_Staff","FullControl","ContainerInherit,ObjectInherit","None","Allow")
$acl2.SetAccessRule($rule2) Set-Acl "C:\Shares\HR_Share" $acl2 # Repeat for
Finance_Share with Finance_Staff group $acl3 = Get-Acl "C:\Shares\Finance_Share"
$acl3.SetAccessRuleProtection($true, $false) $rule3 = New-Object
System.Security.AccessControl.FileSystemAccessRule( "CYBERNOVA\Finance_Staff","FullCont
rol","ContainerInherit,ObjectInherit","None","Allow") $acl3.SetAccessRule($rule3)
Set-Acl "C:\Shares\Finance_Share" $acl3
```

6.4 Verify Share Permissions

```
# View all shares Get-SmbShare | Select-Object Name, Path, Description # View specific
share permissions Get-SmbShareAccess -Name "IT_Share" Get-SmbShareAccess -Name
"HR_Share" Get-SmbShareAccess -Name "Finance_Share" # View NTFS permissions (Get-Acl
"C:\Shares\IT_Share").Access | Select-Object IdentityReference, FileSystemRights
```

■ Test access by logging into a client machine as an HR user and trying to access Finance_Share — access should be denied. Screenshot the denial.

■■ PHASE 7: SECURITY HARDENING

7.1 Enable Windows Firewall

```
# Enable firewall on all profiles Set-NetFirewallProfile -Profile Domain,Public,Private
-Enabled True # Verify firewall status Get-NetFirewallProfile | Select-Object Name,
Enabled
```

7.2 Disable Unnecessary Services

```
# Disable Remote Registry service (security risk) Stop-Service -Name "RemoteRegistry"
-Force Set-Service -Name "RemoteRegistry" -StartupType Disabled # Disable Telnet (if
enabled) Disable-WindowsOptionalFeature -Online -FeatureName TelnetClient -NoRestart #
Disable Print Spooler if not needed (optional) # Stop-Service -Name "Spooler" -Force #
Set-Service -Name "Spooler" -StartupType Disabled # Check running services Get-Service
| Where-Object {$_.Status -eq "Running"} | Select-Object Name, DisplayName
```

7.3 Configure Audit Policies

Auditing tracks logon activity and account changes — critical for security monitoring.

```
# Enable audit policies via auditpol auditpol /set /category:"Logon/Logoff"
/success:enable /failure:enable auditpol /set /category:"Account Logon" /success:enable
/failure:enable auditpol /set /category:"Account Management" /success:enable
/failure:enable auditpol /set /category:"Policy Change" /success:enable /failure:enable
# Verify audit settings auditpol /get /category:*
```

7.4 Secure Administrative Practices

- Rename the default Administrator account

```
Rename-LocalUser -Name "Administrator" -NewName "CyberNova_Admin"
```

- Disable the Guest account

```
Disable-LocalUser -Name "Guest" # Verify Get-LocalUser | Select-Object Name, Enabled
```

- Set minimum password age and enforce history via Default Domain Policy

7.5 Configure Event Log Size

```
# Increase Security event log size to retain more audit logs wevtutil sl Security
/ms:102400000 # View current log settings wevtutil gl Security
```

■ Screenshot the audit policy output and firewall status as evidence.

■ PHASE 8: JOIN CLIENT MACHINES TO DOMAIN

8.1 Configure Client DNS

On each Windows 10/11 client machine, set the DNS server to point to the DC:

```
# On client machine — set DNS to Domain Controller IP # Control Panel > Network >
Adapter Settings > IPv4 Properties # Preferred DNS Server: 192.168.1.10
```

8.2 Join Domain via PowerShell (on Client)

```
# Run on CLIENT01 and CLIENT02 Add-Computer -DomainName "cybernova.local" ` -Credential
(Get-Credential) ` -Restart # When prompted, enter: CYBERNOVA\Administrator and
password
```

8.3 Verify Domain Join (on DC)

```
# On Domain Controller — check joined computers Get-ADComputer -Filter * | Select-Object
Name, DNSHostName, Enabled
```

8.4 Move Computers to Correct OU

```
# Move CLIENT01 to IT Department Computers OU Get-ADComputer "CLIENT01" | Move-ADObject
` -TargetPath "OU=Computers,OU=IT Department,DC=cybernova,DC=local" # Move CLIENT02 to
HR Department Get-ADComputer "CLIENT02" | Move-ADObject ` -TargetPath "OU=HR
Department,DC=cybernova,DC=local"
```

■ Log into a client as a domain user (e.g. aboateng@cybernova.local) and screenshot the successful login.

■ PHASE 9: DOCUMENTATION CHECKLIST

Use this checklist to make sure you have everything before submission.

#	Item	Required Evidence
1	AD DS installed and domain created	Screenshot of Get-ADDomain output
2	4 OUs created (IT, HR, Finance, Remote)	Screenshot of ADUC OU tree
3	12 user accounts created	Screenshot of Get-ADUser output
4	4 security groups created	Screenshot of group members list
5	Password policy configured	Screenshot of policy settings
6	Account lockout policy set	Screenshot of lockout policy
7	Control Panel restricted for HR/Finance	Screenshot of GPO editor setting
8	CMD restricted for non-admins	Screenshot of GPO setting
9	Screen lock policy applied	Screenshot of screensaver GPO
10	IT_Share, HR_Share, Finance_Share created	Screenshot of Get-SmbShare output
11	NTFS permissions set correctly	Screenshot of ACL output
12	Access denied test (wrong dept)	Screenshot of access denied error
13	Firewall enabled	Screenshot of Get-NetFirewallProfile
14	Audit policies configured	Screenshot of auditpol output
15	Client machines joined to domain	Screenshot of domain login on client
16	Network diagram drawn	Diagram in report
17	Technical report written	Word/PDF document

■ QUICK REFERENCE — ALL KEY COMMANDS

AD Installation

```
Install-WindowsFeature -Name AD-Domain-Services -IncludeManagementTools
```

```
Install-ADDSForest -DomainName "cybernova.local" -InstallDns:$true -Force:$true
```

```
Get-ADDomain
```

```
Get-ADDomainController
```

OU Management

```
New-ADOrganizationalUnit -Name "IT Department" -Path "DC=cybernova,DC=local"
```

```
Get-ADOrganizationalUnit -Filter * | Select-Object Name, DistinguishedName
```

User Management

```
New-ADUser -Name "User Name" -SamAccountName "uname" -Enabled $true ...
```

```
Get-ADUser -Filter * | Select-Object Name, SamAccountName, Enabled
```

```
Set-ADAccountPassword -Identity "uname" -Reset -NewPassword (ConvertTo-SecureString
"NewPass@1" -AsPlainText -Force)
```

```
Unlock-ADAccount -Identity "uname"
```

```
Disable-ADAccount -Identity "uname"
```

Group Management

```
New-ADGroup -Name "IT_Admins" -GroupScope Global -GroupCategory Security ...
```

```
Add-ADGroupMember -Identity "IT_Admins" -Members "jmensah","aowusu"
```

```
Get-ADGroupMember -Identity "IT_Admins" | Select-Object Name
```

```
Remove-ADGroupMember -Identity "IT_Admins" -Members "jmensah" -Confirm:$false
```

GPO & Policy

```
gpmc.msc # Open Group Policy Management Console
```

```
gpupdate /force # Force apply all GPOs
```

```
gpresult /r # Show applied GPOs on current machine
```

```
auditpol /get /category:* # View audit policies
```

File Shares

```
New-SmbShare -Name "IT_Share" -Path "C:\Shares\IT_Share" -FullAccess
"CYBERNOVA\IT_Admins"
```

```
Get-SmbShare | Select-Object Name, Path
```

```
Get-SmbShareAccess -Name "IT_Share"
```

```
(Get-Acl "C:\Shares\IT_Share").Access | Select-Object IdentityReference,
FileSystemRights
```

Security

```
Set-NetFirewallProfile -Profile Domain,Public,Private -Enabled True
```

```
Get-NetFirewallProfile | Select-Object Name, Enabled
```

```
Stop-Service -Name "RemoteRegistry" -Force
```

```
auditpol /set /category:"Logon/Logoff" /success:enable /failure:enable
```

```
Rename-LocalUser -Name "Administrator" -NewName "CyberNova_Admin"
```